2024 Spring 生化学 2 English, Home work#6(250507)　　dead line: 23:59, May 14, 2025

学号:

姓名:

Q1. 判断题（请在括号里画√或 X, 1 分/题, 共 25 分）

(1) Glycolysis provides more energy (ATP) from glucose then respiration. (　　)
(2) Pyruvate dehydrogenase is composed of multiple proteins. (　　)
(3) Pyruvate dehydrogenase uses 5 coenzymes (CoA-SH, NADP+, TPP, lipoate, FAD) to convert pyruvate to Acetyl-CoA. (　　)
(4) Lipoic acid bound to Lys. (　　)
(5) Lipoic acid can take 3 forms (Oxidized-, Reduced- and Acetylated-form). (　　)
(6) Synthase catalyze condensation reactions with ATP/GTP. (　　)
(7) Phosphotase removes phosphoryl group from target. (　　)
(8) In the first step of TCA cycle, Acetyl-CoA is reacted with Malate to form Citrate by citrate synthase. (　　)
(9) Aconitase converts citrate to isocitrate, where iron-sulfur center exists in the core of the enzyme. (　　)
(10) When [Iron] is low, aconitase (IPR1) can bind IRE (Iron Response element) of feritin mRNA to regulate the gene expression (moonlighting). (　　)
(11) Aconitase is the only example of protein moonlighting. (　　)
(12) α-Ketoglutarate dehydrogenase complex converts a-ketoglutarate to Succinyl-CoA using several cofactors such as CoA-SH, NAD+,TPP, lipoate and FAD. (　　)
(13) Succinyl-CoA synthetase uses GTP to convert Succinyl-CoA to Succinate. (　　)
(14) Succinate dehydrogenase uses NAD+ to convert Succinate to Fumarate. (　　)
(15) Malate dehydrogenase uses NAD+ to convert Malate to Oxaloacetate. (　　)
(16) 1 round of TCA cycle obtain 2x NADH, 1x $FADH_2$, 1x GTP (ATP), and release 2x CO_2. (　　)
(17) For most of the TCA cycle step, physiological free energy change (ΔG) is largely negative value. (　　)
(18) In plants, acetic acid can be converted to Oxaloacetate by Glyoxylate cycle. (　　)
(19) Not-Respiratory Organisms has α-Ketoglutarate dehydrogenase. (　　)

(20) In Glyoxylate cycle, Isocitrate is converted to Malate by two enzymes called Isocitrate lyase and Malate synthase. (　　)

(21) In Glyoxylate cycle, FAD is used in addition to NAD+. (　　)

(22) Animal also has Malate synthase. (　　)

(23) ATP upregulates the activity of pyruvate dehydrogenase complex. (　　)

(24) Succinyl-CoA downregulates the activity of α-Ketoglutarate dehydrogenase complex. (　　)

(25) NADH upregulates the activity of citrate synthase. (　　)

Q2. Write down the 5 steps of oxidative decarboxylation of PDH (words or pictures). (各 1 分、共 5 分)

Q3. Draw TCA cycle (共 10 分)

Q4. Draw Glyoxylate cycle (共 10 分)